Unit-2

Topics:

- **Set Theory**
- **Counting & Combinatorics**
- **Recurrence Relations**

◆ PART 1: SET THEORY

Video Based Learning : https://youtu.be/enHai1y5Knc?si=mbY58NNLlUzyHJmh/

1 Set

✅ **Definition :** A set is a well-defined collection of distinct objects.

Understanding

- "Well-defined" means no confusion.
- Objects are called **elements** or **members**.
- Symbol for belonging: ∈

Example:

A = {1, 2, 3}

1 ∈ A

5 ∉ A

2 Representation of Sets

Sets can be represented in two main methods:

1. Roster (Tabular) Method
2. Set Builder Method

3 Roster Method (Tabular Form)

✅ **Definition :** In the roster method, all elements of a set are listed inside curly brackets separated by commas.

Example

A = {2, 4, 6, 8}

Rules:

- Elements written inside { }
- No repetition
- Order does not matter

Example:

{1,2,3} = {3,2,1}

4 Set Builder Method

✅ **Definition :** In set builder method, a set is described by stating a property that its elements satisfy.

General Form : A = {x | property of x}

Read as: "A is the set of all x such that x satisfies the given property."

Example

A = {x | x is even natural number less than 10}

This means:

A = {2, 4, 6, 8}

5 Empty Set (Null Set)

✅ **Definition :** A set that contains no elements is called an empty set or null set.

Symbol : ∅ or { }

Example

A = {x | x is a natural number less than 0}

No such number → So A = ∅

6 Cardinality of a Set

✅ **Definition :** The cardinality of a set is the number of elements in the set.
Notation: $|A|$ = number of elements in set A
Example
A = {1, 2, 3}
$|A| = 3$
For empty set:
$|\varnothing| = 0$
7 Popular Sets in Mathematics
These are commonly used sets:

Symbol	Set Name	Elements
$\mathbb{N}$	Natural Numbers	{1,2,3,...}
$\mathbb{Z}$	Integers	{...,-2,-1,0,1,2,...}
$\mathbb{Q}$	Rational Numbers	Numbers of form p/q
$\mathbb{R}$	Real Numbers	All rational & irrational numbers
$\mathbb{C}$	Complex Numbers	a + bi

8 Finite Set
✅ **Definition :** A set that contains a limited number of elements is called a finite set.
Example
A = {1,2,3,4}
$|A| = 4$

9 Infinite Set
✅ **Definition :** A set that contains unlimited or infinitely many elements is called an infinite set.
Example
$\mathbb{N}$ = {1,2,3,...}
No ending → Infinite

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10 Equal Sets
✅ **Definition :** Two sets A and B are said to be equal if they contain exactly the same elements.
Condition
A = B if and only if

- Every element of A is in B
- Every element of B is in A

Example
A = {1,2,3}
B = {3,2,1}
A = B
Order does not matter.

12 Subset
✅ **Definition :** A set A is called a subset of set B if every element of A is also an element of B.
Notation
$A \subseteq B$
Example
A = {1,2}
B = {1,2,3,4}
$A \subseteq B$

13 Proper Subset
✅ **Definition :** set A is called a proper subset of B if $A \subseteq B$ and $A \neq B$.
Notation
$A \subset B$
Example

A = {1,2}
B = {1,2,3}
$A \subset B$

14 Superset
✅ **Definition :** A set B is called a superset of A if all elements of A are contained in B.
Notation
$B \supseteq A$

15 Universal Set
✅ **Definition :** The universal set is the set that contains all elements under consideration in a particular discussion.
Notation
U
Example:
If discussion is about natural numbers less than 10:
U = {1,2,3,4,5,6,7,8,9}

16 Complement of a Set
✅ **Definition:** The complement of a set A is the set of all elements in the universal set U that are not in A.
Notation
A' or A^c
Formula
A' = U − A

17 Singleton Set
✅ **Definition:** A set containing exactly one element is called a singleton set.
Example
A = {5}
|A| = 1

18 Power Set
✅ **Definition:** The power set of A is the set of all subsets of A.
Notation
P(A)
Formula
If |A| = n
Number of subsets = 2^n
Example
A = {1,2}
P(A) = { ∅, {1}, {2}, {1,2} }
Total subsets = 4 = 2^2

19 Open and Closed Interval (Real Number Sets)
Used mainly in $\mathbb{R}$.

Open Interval
Definition: (a, b) represents all real numbers between a and b but excluding a and b.
Example:
(1,5) = {x | 1 < x < 5}

Closed Interval
Definition: [a, b] represents all real numbers between a and b including a and b.
Example:

[1,5] = {x | 1 ≤ x ≤ 5}

Mixed Interval
[a, b) → includes a, excludes b
(a, b] → excludes a, includes b

1 Inclusion and Exclusion Principle (Sets)
For Two Sets
$|A \cup B| = |A| + |B| - |A \cap B|$
Reason:
Intersection counted twice → subtract once.

For Three Sets
$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$

2 Venn Diagram
Definition
A Venn diagram is a pictorial representation of sets using closed curves to show relationships between sets.

Uses
- Show union
- Show intersection
- Show complement
- Solve inclusion-exclusion problems

Set Operations

3 Union of Sets (∪)
✅ **Definition :** The union of two sets A and B is the set of all elements that belong to A or B or both.
Notation
$A \cup B$
Mathematical Form
$A \cup B = \{x \mid x \in A \ \text{or} \ x \in B\}$
Example
A = {1,2,3}
B = {3,4,5}
A ∪ B = {1,2,3,4,5}

◆ **Properties of Union**

Property	Law
Commutative	A ∪ B = B ∪ A
Associative	(A ∪ B) ∪ C = A ∪ (B ∪ C)
Identity	A ∪ ∅ = A
Domination	A ∪ U = U
Idempotent	A ∪ A = A

3 Intersection of Sets (∩)
✅ **Definition :** The intersection of two sets A and B is the set of elements common to both A and B.
Notation
$A \cap B$
Mathematical Form
$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
Example
A ∩ B = {3}

- **Properties of Intersection**

Property	Law
Commutative	A ∩ B = B ∩ A
Associative	(A ∩ B) ∩ C = A ∩ (B ∩ C)
Identity	A ∩ U = A
Domination	A ∩ ∅ = ∅
Idempotent	A ∩ A = A

4 Set Difference (−)
✅ **Definition:** The difference of sets A and B is the set of elements that belong to A but not to B.
Notation
A − B
Mathematical Form
A − B = {x | x ∈ A and x ∉ B}
A-B = A − (A ∩ B)
Example:
A={1,2,3,4},
B={3,4,5,6,7}
A-B={1,2}

5 Symmetric Difference (Δ)
✅ **Definition :** The symmetric difference of A and B is the set of elements which belong to either A or B but not both.
Notation
A Δ B or A ⊕ B
Formula
A Δ B = (A − B) ∪ (B − A)
A⊕B =(A-B) ∪ (B-A)
Example
A = {1,2,3}
B = {3,4,5}
A-B={1,2}
A Δ B = {1,2,4,5}

- **Properties of Symmetric Difference**

Property	Law
Commutative	A Δ B = B Δ A
Associative	(A Δ B) Δ C = A Δ (B Δ C)
Identity	A Δ ∅ = A
Self Law	A Δ A = ∅

6 Cartesian Product (×)
✅ **Definition :** The Cartesian product of A and B is the set of all ordered pairs (a,b) where a ∈ A and b ∈ B.
Notation
A × B
Mathematical Form

$$A \times B = (a,b) \mid a \in A, b \in B$$

Example
A = {1,2}
B = {3,4}
A × B = {(1,3),(1,4),(2,3),(2,4)}

◆ **Important Property**

Property	Statement
Not Commutative	$A \times B \neq B \times A$
Cardinality	$\|A\times B\| = \|B \times A\|$
Empty Set Property	$A\times \varnothing = \varnothing$ and $\varnothing\times A=\varnothing$
Subset Property	If $A\subseteq B$, then $A\times C \subseteq B\times C$
Distributive over Union	$A\times(B\cup C) = (A\times B)\cup(A\times C)$
Distributive over Intersection	$A\times(B\cap C) = (A\times B)\cap(A\times C)$

7 Disjoint Sets

✅ **Definition :** Two sets A and B are said to be disjoint if they have no elements in common.

Condition

[$A \cap B = \varnothing$]

Example

$A = \{1,2\}$

$B = \{3,4\}$

$A \cap B = \varnothing$ → Disjoint

8 Complement of Set

Definition: Complement of A is the set of elements in universal set U but not in A.

[$A' = U - A$]

9 Basic Set Identities (Very Important for Exam)

Identity Laws

$A \cup \varnothing = A$

$A \cap U = A$

Domination Laws

$A \cup U = U$

$A \cap \varnothing = \varnothing$

Complement Laws

$A \cup A' = U$

$A \cap A' = \varnothing$

De Morgan's Laws

[$(A \cup B)' = A' \cap B'$]

[$(A \cap B)' = A' \cup B'$]

Distributive Laws

[$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$]

[$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$]

Absorption Laws

$A \cup (A \cap B) = A$

$A \cap (A \cup B) = A$

📊 **Quick Comparison Table**

Operation	Symbol	Meaning

Union	∪	All elements
Intersection	∩	Common elements
Difference	−	In A but not in B
Symmetric Difference	Δ	In either but not both
Cartesian Product	×	Ordered pairs
Complement	A'	Outside A
Disjoint	A ∩ B = ∅	No common elements

◆ 🔢 Part : 2 - Counting & Combinatorics

Video Based Learning : https://youtu.be/yurSCPPloMs?si=Yb-6vsnwShu51iaO/ or
Video Based Learning 2: https://youtu.be/Tr-TVt5JAWY?si=hRu8Hxn-WqhQTpUw/

1 Basic Counting Principle (Fundamental Principle of Counting)

✔ **Definition:** If one task can be performed in m ways and another independent task can be performed in n ways, then both tasks together can be performed in m × n ways.
This is also called the **Multiplication Principle**.

💭 Understanding

If:

- First choice → m options
- Second choice → n options

Total combinations = m × n

◆ Example 1

You have:

- 3 shirts
- 2 pants

How many different outfits?
Answer:
3 × 2 = 6 outfits

◆ Example 2

A password has:

- 4 letters
- 3 digits

Total possible passwords?
4 × 3 = 12

◆ Extension of Basic Counting Principle

If there are multiple tasks:

- Task 1 → n_1 ways
- Task 2 → n_2 ways
- Task 3 → n_3 ways

Total ways:
[$n_1 \times n_2 \times n_3$]

◆ Example

A code consists of:

- 2 letters (each 26 choices)
- 3 digits (each 10 choices)

Total codes:
[26 × 26 × 10 × 10 × 10]

2 Factorial

✅ **Definition (Exam Ready) :** The factorial of a positive integer n is the product of all positive integers from 1 to n.

Notation

n!

Formula

[$n! = n \times (n-1) \times (n-2) \times \dots \times 1$]

Examples

1! = 1
2! = 2 × 1 = 2
3! = 3 × 2 × 1 = 6
4! = 4 × 3 × 2 × 1 = 24
5! = 120

Important Property

[$n! = n \times (n-1)!$]
Example:
5! = 5 × 4!

Special Case

[0! = 1]
Very important in combinations.

◆ Why Factorial is Important?

Factorial is used in:

- Permutations (nPr)
- Combinations (nCr)
- Arrangements
- Probability

◆ Addition Principle

✅ **Definition :** If one task can be performed in m ways and another task can be performed in n ways, and the two tasks cannot occur simultaneously, then the total number of ways of performing either task is m + n.
This is also called the **Sum Rule**.

🧠 Key Condition

The tasks must be **mutually exclusive** (no overlap).
Meaning:
You choose either task 1 or task 2 — not both.

◆ Example 1

A student can choose:

- 3 mathematics books
- 2 physics books

If he selects **one book**, how many choices?
Total = 3 + 2 = 5
Why addition?
Because he chooses either math OR physics.

◆ Example 2

A person can travel:

- By bus (4 routes)
- By train (3 routes)

Total travel options:
4 + 3 = 7

◆ Compare Addition vs Multiplication Principle

Situation	Use
Choose one option	Addition
Perform multiple tasks together	Multiplication

◆ Clear Difference Example

If:

- Choose one shirt (3 options)
- Choose one pant (2 options)

To form outfit → 3 × 2 = 6 (Multiplication)
If:
Choose either a shirt OR a pant → 3 + 2 = 5 (Addition)

◆ Set Theory Connection

If A and B are disjoint sets:
$|A \cup B| = |A| + |B|$
If not disjoint:
$|A \cup B| = |A| + |B| - |A \cap B|$
(This becomes Inclusion-Exclusion Principle)

◆1 What is Permutation?

✔ **Definition:** A permutation is an arrangement of objects in which the order of arrangement is important.
Key point: **Order matters.**

◆2 Formula of Permutation

If we have:

- n distinct objects
- We want to arrange r objects

Then number of permutations is:

$$\mathbf{nPr} = \frac{n!}{(n-r)!}$$

3 Why This Formula?

To arrange r objects from n:
First position → n choices
Second position → (n − 1) choices
Third position → (n − 2) choices
...
Total:

$$n \times (n-1) \times (n-2) \times \ldots (r \text{ terms})$$

Which equals:

$$\frac{n!}{(n-r)!}$$

4 Example 1

Arrange 2 letters from A, B, C.
Here:
n = 5
r = 3

$$5P3 = \frac{5!}{(5-3)!} = \frac{5!}{2!} = \frac{120}{2} = 60$$

Possible arrangements:
AB, BA, AC, CA, BC, CB
Notice:
AB ≠ BA
Order changes → different permutation.

5 Example 2
How many ways can 3 students be arranged in a row from 5 students?
n = 3
r = 2

$$3P2 = \frac{3!}{(3-2)!} = \frac{3!}{1!} = 6$$

6 Special Case: When r = n

$$nPn = n!$$

Meaning:
Arranging all objects.
Example:
Arrange 4 books in a row:
4! = 24 ways

Difference Between Permutation and Combination

Basis	**Permutation**	**Combination**
Order	Important	Not important
Formula	nPr	nCr
Example	Seat arrangement	Team selection

Types of Permutations (Important Concept)
(A) Permutation Without Repetition
Each object used once.
Formula:

$$nPr = \frac{n!}{(n-r)!}$$

(B) Permutation With Repetition
If repetition allowed:
Each position has n choices.
Total:

$$n^r$$

Example:
4-digit PIN using digits 0–9
10 × 10 × 10 × 10
= 10^4
= 10000

Important Properties

1. nP0 = 1
2. nP1 = n
3. nPr = nP(n−r) × (n−r)!

Exam-Type Questions

1. Arrange word "CAT"
2. How many 3-letter words from 5 letters?

3. Number of 4-digit numbers from digits 1–9 without repetition
4. Number of ways to arrange 6 people in a row

◆ What is Circular Permutation?

✅ **Definition :** A circular permutation is an arrangement of objects in a circle where the relative position matters but the starting point does not.

Key idea:

In a circle, there is **no fixed starting position**.

◆ Why Formula Changes in Circle?

In linear arrangement:

$$n! \text{ ways}$$

In circular arrangement:

Rotating the circle does not create a new arrangement.

Example:

ABC, BCA, CAB (in circle) are same.

So we divide by n.

◆ Formula of Circular Permutation

For n distinct objects arranged in a circle:

$$(n-1)!$$

◆ Why (n − 1)! ?

Because:

Fix one object to remove rotation effect.

Then arrange remaining (n − 1) objects.

So:

$$(n-1)!$$

◆ Example 1

Arrange 4 people around a round table.

$$(4-1)!=3!=6$$

◆ Example 2

Arrange 6 beads in a circular necklace.

$$(6-1)!=5!=120$$

◆ Important Case: Necklace (Clockwise & Anticlockwise Same)

If reflections are also identical (like necklace):

Formula becomes:

$$\frac{(n-1)!}{2}$$

Because clockwise and anticlockwise look same.

◆ Comparison Table

Type	Formula
Linear Permutation	n!
Circular Permutation (round table)	(n − 1)!
Necklace (reflection same)	(n − 1)! / 2

◆ Special Case

If 2 objects: (2 − 1)! = 1

Only 1 circular arrangement.

◆ **Typical Questions**

1. 5 persons sit around a round table.
2. 8 people sit in circular order.
3. 6 beads form necklace.

🎯 **Common Mistake**

Students use n! instead of (n−1)!
Always remember:
Circle removes one degree of freedom.

Permutation Formulas Summary Table

Formula Name	Formula	Description	Explanation	Example
Permutation of n objects (all arranged)	$\mathbf{nPn} = n!$	Number of ways to arrange **n distinct objects** taking **all objects at a time**	When every object must be used and order matters.	Arrange A, B, C → (3! = 6) ways: ABC, ACB, BAC, BCA, CAB, CBA
Permutation of n objects taken r at a time	$\mathbf{nPr} = \frac{n!}{(n-r)!}$	Number of ways to select and arrange **r objects from n distinct objects**	First choose r objects, then arrange them. Order matters.	From {A,B,C,D}, arrange 2 → ($4P2 = \frac{4!}{2!} = 12$)
Permutation when all objects are not distinct (repeated objects)([youtube]**)**	$\frac{\boldsymbol{n!}}{\boldsymbol{n1!} \times \boldsymbol{n2!}}$	Used when some objects are identical	Divide by factorial of repeated items because identical objects produce duplicate arrangements	Word **BALLOON** (7 letters, L repeated 2 times, O repeated 2 times) → ($\frac{7!}{2!2!}$ = 1260)
Permutation with repeated objects allowed	n^r	Each position can be filled with any of the n objects repeatedly	Objects can repeat in arrangement	Form 3-digit numbers using {1,2,3} → (3^3 = 27)
Circular Permutation (different objects)	$(n-1)!$	Arrangement of objects around a **circle**	In a circle, rotations are considered same, so fix one position and arrange remaining	5 people around a round table → ((5-1)! = 24)
Circular Permutation (necklace or bracelet)	$\frac{(\boldsymbol{n}-\mathbf{1})!}{\mathbf{2}}$	Circular arrangement where **clockwise and anticlockwise are same**	Used in necklaces or garlands	6 beads in necklace → (\frac{5!}{2} = 60)

Quick Recall (Exam Ready)

Case	Formula
Arrange n distinct objects	$\mathbf{nPn} = n!$
Arrange r objects from n	$\mathbf{nPr} = \frac{n!}{(n-r)!}$
Objects repeated	$\frac{\boldsymbol{n!}}{\boldsymbol{n1!} \times \boldsymbol{n2!}}$
Repetition allowed	n^r
Circular permutation	$(n-1)!$

Circular (necklace/garland)	$\frac{(n-1)!}{2}$

Important Notes for Exams

1. **Permutation always cares about order.**
2. **Factorial definition**
   n! = n(n-1)(n-2)...1
3. If **objects are identical**, divide by factorial of repetitions.
4. In **circular permutation**, one position is fixed to avoid counting rotations.

If useful, a **second table comparing Permutation vs Combination formulas** can also be provided for faster recall in exams.

1 What is Combination?

✅ **Definition :** A combination is the selection of objects in which the order of selection is not important.

Key idea:

Order does NOT matter.

Formula of Combination

If we have:

- n distinct objects
- We select r objects

Then number of combinations:

$$nCr = \frac{n!}{r!\,(n-r)\,!}$$

Why This Formula?

We know:

$$nPr = \frac{n!}{(n-r)\,!}$$

But in permutations, order matters.

Each group of r objects can be arranged in r! ways.

So to remove ordering:

$$nCr = \frac{nPr}{r!}$$

Which becomes:

$$\frac{n!}{r!(n-r)!}$$

Example 1

Choose 2 students from 4 students.

n = 4

r = 2

$$4C2 = \frac{4!}{2!\ (4-2)\,!} = \frac{4!}{2!\,2!} = \frac{24}{4} = 6$$

Possible selections:

AB, AC, AD, BC, BD, CD

Notice:

AB and BA are same → order not counted.

Example 2

Choose 3 books from 5 books.

$$5C3 = \frac{5!}{3!\ (5-3)\,!} = \frac{5!}{3!\,2!} = \frac{120}{12} = 10$$

Important Properties

1 Symmetry Property

$$nCr = nC(n-r)$$

Example:
5C2 = 5C3

2 Special Values

$$nC0=1$$
$$nCn = 1$$
$$nC1= n$$

Difference: Permutation vs Combination

Basis	Permutation (nPr)	Combination (nCr)
Order	Important	Not important
Formula	n!/(n-r)!	n!/(r!(n-r)!)
Example	Seating arrangement	Team selection

When to Use Combination?
Use when:

- Selecting committee
- Forming team
- Choosing objects
- Selecting questions

If arrangement/position not asked → use nCr.

◆ **Practical Examples**

1. Select 3 subjects out of 7 → 7C3
2. Choose 4 players from 10 → 10C4
3. Select 2 questions from 5 → 5C2

Relationship Between nPr and nCr

$$nPr = nCr \times r!$$

◆ **Inclusion–Exclusion Principle**
This is used to **avoid double counting**.

◆ **Inclusion–Exclusion Principle for Two Sets**
✅ **Definition:** The Inclusion–Exclusion Principle states that for any two finite sets A and B, the number of elements in their union is given by:
$|A \cup B| = |A| + |B| - |A \cap B|$

💭 **Why Do We Subtract?**
When we add |A| + |B|:
Common elements (A ∩ B) are counted twice.
So we subtract once.

◆ **Example 1**
In a class of 50 students:
30 like Mathematics
25 like Physics
10 like both
Find students who like at least one subject.
$|M \cup P| = |M| + |P| - |M \cap P|$ =30+25−10=45= 30 + 25 - 10 = 45=30+25−10=45
Answer: 45 students

◆ **Inclusion–Exclusion Principle for Three Sets**

✅ **Formula**

$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$

💭 **Why Add Back the Triple Intersection?**

Because:
When subtracting pairwise intersections,
the common part of all three gets removed three times.
So we add it back once.

◆ **Example 2**

In a survey of 100 people:
50 like Tea
40 like Coffee
30 like Juice
15 like Tea & Coffee
10 like Coffee & Juice
5 like Tea & Juice
3 like all three
Find people who like at least one.
$|T \cup C \cup J| = 50+40+30-15-10-5+3$
$=120-30+3 = 93$
Answer: 93 people

◆ **General Idea (n Sets)**

For n sets:

- Add all individual sizes
- Subtract all pairwise intersections
- Add triple intersections
- Subtract quadruple intersections
- Continue alternating signs

◆ **Special Case: Disjoint Sets**

If A and B are disjoint: $|A \cap B|=0$
So: $|A \cup B| = |A| + |B|$

◆ **Real-Life Applications**

✔ Counting students
✔ Survey problems
✔ Probability
✔ Venn diagram problems
✔ Avoiding double counting

◆ **Venn Diagram Interpretation**

- Each overlapping region represents intersection.
- Use formula to fill missing region values.

◆ **What is a Binomial?**

Video Based Learning : https://youtu.be/j9GohkxkjGk?si=eG7irSA_oXO_7NQ3/

✅ **Definition (Exam Ready)**

A binomial is an algebraic expression containing exactly two terms connected by addition or subtraction.

Examples

- a + b

- x – y
- 2p + 3q

"Bi" means two.
Binomial = two terms.

◆ Expanding a Binomial (Basic Idea)
Consider:

$$(a+b)^2$$

Expand:

$$(a+b)(a+b)$$

Result:

$$a^2 + 2ab + b^2$$

Now:

$$(a+b)^3$$

Expand:

$$(a+b)(a+b)(a+b)$$

Result:

$$a^3 + 3a^2\mathrm{b} + 3ab^2 + b^3$$

◆ Pattern of Coefficients
Look at expansions:

Power	Expansion	Coefficients
$(a+b)^1$	a + b	1 1
$(a+b)^2$	$a^2 + 2ab + b^2$	1 2 1
$(a+b)^3$	$a^3 + 3a^2b + 3ab^2 + b^3$	1 3 3 1
$(a+b)^4$	$a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$	1 4 6 4 1

Notice pattern:
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
This is **Pascal's Triangle**.

◆ Binomial Coefficient
✐ **Definition:** The binomial coefficient is the coefficient of a term in the expansion of a binomial expression.
Notation:

$$\binom{n}{r} \quad \text{or} \quad nCr$$

Formula:

$$\binom{n}{r} = \frac{n!}{r!\,(n-r)\,!}$$

Example
In:

$$(a+b)^4$$

Coefficient of a^2b^2 is:

$$\binom{4}{2} = 6$$

◆ Pascal's Triangle
Construction Rule
Each number is sum of two numbers above it.
Example:

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

Mathematically:

$$nCr = (n-1)\,C\,(r-1) + (n-1)\,Cr$$

◆ Binomial Theorem

✅ Statement (Exam Ready)

The Binomial Theorem states that:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

What It Means

Each term has:

- Coefficient → nCr
- Power of a → decreases
- Power of b → increases

◆ Structure of General Term

The $(r+1)^{th}$ term:

$$\boldsymbol{T_{r+1} = \binom{n}{r} a^{n-r} b^r}$$

◆ Example

Expand:

$$\boldsymbol{(a+b)^3}$$

Using formula:

$$= \binom{3}{0} \boldsymbol{a^3} + \binom{3}{1} \boldsymbol{a^2 b} + \binom{3}{2} \boldsymbol{ab^2} + \binom{3}{3} \boldsymbol{b^3}$$
$$= \boldsymbol{1a^3 + 3a^2b + 3ab^2 + 1b^3}$$

◆ Important Observations

1. Total number of terms in expansion = index count + 1 ($(a+b)^6$ = 7 terms)
2. Power of the first quantity 'a' go to decreasing by 1 (start from a^n and end with a^0)
3. Power of the second quantity 'b' go to increasing by 1 (start from b^0 and end with b^n)
4. Sum of coefficients = 2^n
5. Middle coefficient largest
6. Coefficients symmetric

Example:

5C2 = 5C3

◆ Why This Topic Is Important?

✔ Used in probability
✔ Used in combinatorics
✔ Used in algebra expansion
✔ Appears in GTU exams regularly

◆ What is Pigeonhole Principle?

✅ Basic Definition: If n objects are placed into m boxes and n > m, then at least one box contains more than one object.

In simple words:

If items are more than boxes → some box must contain at least 2 items.

◆ Why It Is Called Pigeonhole?

Imagine:

- Pigeons = objects
- Holes = boxes

If 5 pigeons enter 4 holes,
at least one hole has 2 pigeons.

◆ Simple Example

There are 13 students.
There are 12 months in a year.
Prove that at least two students have birthday in same month.
Here:
Objects = 13 students
Boxes = 12 months
Since 13 > 12
→ At least one month has 2 students.

◆ General (Strong) Pigeonhole Principle

✅ **Definition:** If n objects are placed into m boxes, then at least one box contains at least:

$$\left\lceil \frac{n}{m} \right\rceil$$

objects.
(ceiling function means round up)

◆ Example of Strong Form

Distribute 25 balls into 4 boxes.
Minimum balls in one box?

$$\left\lceil \frac{25}{4} \right\rceil = \lceil 6.25 \rceil = 7$$

So at least one box has 7 balls.

◆ Typical Exam Questions

✔ Show that in group of 6 people, at least 2 have same number of friends.
✔ Show that among 51 numbers, at least 2 have same remainder when divided by 50.
✔ Show that in any 367 people, at least 2 share same birthday.

◆ Example (Remainder Type)

Choose 6 numbers from 1 to 10.
Prove at least 2 numbers have same remainder when divided by 5.
Possible remainders when divided by 5:
0,1,2,3,4 → 5 possible remainders
6 numbers → 5 boxes
Since 6 > 5
→ At least two numbers share remainder.

◆ How to Solve Pigeonhole Problems

Step 1: Identify objects
Step 2: Identify boxes
Step 3: Compare n and m
Step 4: Apply principle

◆ Important Insight

Pigeonhole principle guarantees existence.
It does not tell which box — only that one must exist.

◆ **Summary**

Concept	Meaning
Basic Form	If n > m → one box has ≥ 2
Strong Form	At least ⌈n/m⌉ items in one box
Used in	Proof problems

Video on pigeonhole principle: https://youtu.be/B2A2pGrDG8I?si=J7siMe6vsf801-7Y/

Recurrence Relations:

Video based learning : https://youtu.be/2ii8mCFXj88?si=sOjaWQhT6QJEyOgj/

1 What is a Recurrence Relation?

Definition : A recurrence relation is an equation that defines each term of a sequence using one or more of the previous terms.

Simple Idea : Each value depends on earlier values.
Example:

$$a_n = a_{n-1} + 2$$

Meaning:
Each term = previous term + 2.

Example Sequence
If

$$a_1 = 3$$

Then

n	Formula	Result
a_1	given	3
a_2	$a_1 + 2$	5
a_3	$a_2 + 2$	7
a_4	$a_3 + 2$	9

Sequence:
3, 5, 7, 9, …

2 Sequence
Definition: A sequence is an ordered list of numbers.
Example:
2, 4, 6, 8, 10
Here each number is called a **term**.
Notation:

$$a_1, a_2, a_3, \ldots, a_n$$

3 Initial Conditions
Definition: Initial conditions are the starting values needed to generate the sequence.
Without them we cannot calculate later terms.

Example

$$a_n = a_{n-1} + 2$$

Needs starting value.

Example:

$$a_1 = 3$$

4 Order of Recurrence Relation

Definition : The order of a recurrence relation is the number of previous terms used to define the next term.

Example 1

$$a_n = a_{n-1} + 3$$

Order = 1
(uses one previous term)

Example 2

$$a_n = a_{n-1} + a_{n-2}$$

Order = 2
(uses two previous terms)

5 Fibonacci Sequence (Famous Example)

Recurrence relation:

$$F_n = F_{n-1} + F_{n-2}$$

Initial conditions:

$$F_0 = 0, \quad F_1 = 1$$

Sequence:
0, 1, 1, 2, 3, 5, 8, 13, ...
Each term = sum of previous two terms.

6 Types of Recurrence Relations (Basic Idea)

Linear Recurrence

Example:

$$a_n = 2a_{n-1} + 3$$

Variables appear linearly.

Homogeneous Recurrence

Example:

$$a_n = 2a_{n-1}$$

No constant term.

Non-Homogeneous Recurrence

Example:

$$a_n = 2a_{n-1} + 5$$

Contains constant term.

7 Why Recurrence Relations Are Important

Used in:

- Algorithm analysis
- Dynamic programming
- Fibonacci problems
- Computer science complexity

Example:
Merge sort time complexity uses recurrence.

8 What Usually Comes in Exam

✔ Definition of recurrence relation
✔ Find sequence from recurrence
✔ Identify order of recurrence
✔ Fibonacci example
Complex solving usually comes in later topics.